
VELA: Stationary Latent Targets for Label-Efficient Strong-Lensing Representations

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Abstract

Strong-lensing surveys provide many images but few trustworthy labels, making frozen representation quality especially important. We study self-supervised pretraining on DeepLenseSim with a common 21.5M-parameter encoder and a repaired evaluation protocol. VELA predicts latent targets from a small scaffold trained once and then frozen. Across three seeds, VELA reaches 0.900 ± 0.003 frozen linear accuracy with all downstream labels, 0.815 ± 0.011 with 10%, and $R^2 = 0.821 \pm 0.012$ for axion-mass regression. Relative to MAE, the strongest baseline, VELA gains 6.9 accuracy points with all labels, 11.7 points with 10%, and 0.125 in R^2 . The advantage remains when I-JEPA receives the same visible-token budget as VELA. Controls qualify the mechanism: cumulative low-pass and residual-band targets perform similarly, while an untrained scaffold or a single supervised depth succeeds in two seeds but becomes unreliable in the third. The evidence supports stationary latent targets as a useful small-data design choice, but does not isolate stationarity as the sole cause of VELA’s gain.

1 Introduction

Strong gravitational lenses probe dark substructure [14]. Wave-like dark matter provides an alternative to conventional cold dark matter [10], and simulated substructure morphologies can be distinguished with neural networks [1]. Simulated labels inherit the simulator’s assumptions, while real images lack dark-matter labels. Self-supervision has been used for lens discovery [12], lens representations with physically motivated augmentations [7], and MAE pretraining for dark-matter classification and super-resolution [11].

Masked image modelling offers several target choices. MAE reconstructs fixed pixels [9]; I-JEPA predicts latents from an exponential-moving-average teacher that evolves during training [2]. Fixed learned targets also have precedent: BEiT predicts discrete codes from a pretrained tokenizer [4].

VELA predicts continuous multiscale latents from a small reconstruction scaffold trained on the lens images and then frozen. A corruption-conditioned predictor maps the encoder’s visible tokens to these targets. Only the encoder is retained for downstream evaluation, using the same architecture as the baselines. We test this design where labels are scarce and the signal is faint and instrument-dependent.

We compare VELA with MAE, I-JEPA, context-matched I-JEPA, and a SIGReg-based JEPA adaptation on DeepLenseSim. The protocol corrects four defects in an earlier pilot: unequal visible-token budgets, an unshuffled I-JEPA loader, checkpoint selection on the test set, and checkpoint reuse after configuration changes. We report all three seeds and controls for target decomposition, scaffold training, and depth ladder.

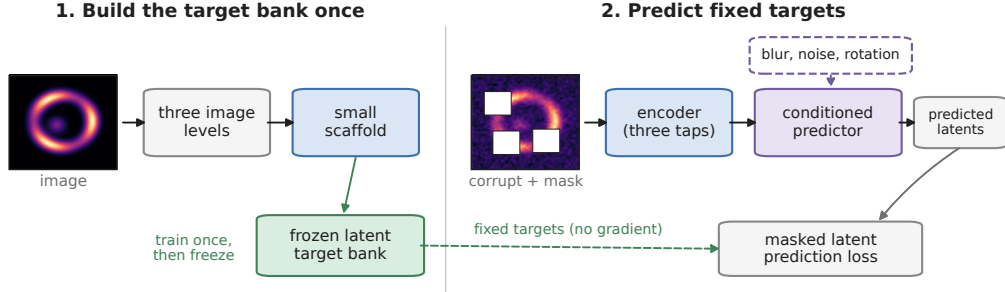


Figure 1: VELA has a target-building stage and an encoder-training stage. The scaffold is trained on a three-level image decomposition and then frozen. During encoder pretraining, the instrument operator predicts the resulting fixed latents from corrupted, partially observed inputs. Only the encoder is used in downstream evaluation.

2 VELA

Fixed multiscale targets. For an input image x , we construct either a cumulative low-pass ladder $[x, G_3x, G_{10}x]$ or residual bands $[x - G_3x, G_3x - G_{10}x, G_{10}x]$, where G_σ denotes Gaussian smoothing. A four-block, 128-dimensional scaffold is trained for 30 epochs to reconstruct masked patches from these three levels. We then freeze the scaffold and precompute its patch latents. Whitening is fitted separately for each level. Once this stage is complete, the targets are independent of subsequent encoder updates.

Encoder and instrument operator. The encoder is a three-block ViT [8] with width 768, 12 heads, and 15×15 patches, giving 100 tokens for a 150×150 image and 21.52M parameters. Its three blocks are tapped at successive depths. Before masking, an image is perturbed by Gaussian blur, additive noise, a multiple-of- 90° rotation, and an optional resolution change. The encoder observes 50 tokens. A two-block cross-attention predictor receives the corruption parameters and predicts the scaffold target at masked locations.

The predictor includes a diagonal channel $M(c) = \exp(\{\text{diag}(Ac)\})$ for additive blur-variance coordinates. This channel is composable by construction, but the complete predictor also contains learned residual blocks. The present experiments evaluate the encoder representation, not long-horizon composition or inversion of the predictor.

Objective. Let z^ℓ be the encoder tokens at depth ℓ , g_ℓ the corresponding predictor, and $\tilde{T}^\ell$ the whitened frozen target. The main term is a precision-weighted Smooth L1 loss on masked patches,

$$\mathcal{L}_{\text{pred}} = \frac{1}{3} \sum_{\ell=1}^3 \mathbb{E}_{i \in \mathcal{M}} \rho(g_\ell(z^\ell, a)_i, \tilde{T}_i^\ell). \quad (1)$$

The implementation also uses lower-weight Gram, inverse-corruption, isotropy, cross-level decorrelation, forward-prediction, and two-step composition terms. These terms are held fixed across the two full VELA variants. Our experiments do not separately attribute their effects.

3 Experimental design

Data and splits. We use Michael W. Toomey’s DeepLenseSim repository [13], whose simulations use `lenstronomy` [5, 6]. We select 9,999 Model I images for unlabelled pretraining and 9,000 Model II images from three balanced classes—axion, cold dark matter (CDM), and no substructure—split into 5,760 train, 1,440 validation, and 1,800 test examples. Model I uses 150×150 images; Model II uses 64×64 images with Euclid-like observing conditions, resized bilinearly to 150×150 here. Images are individually min–max normalized. The axion subset includes mass metadata for regression.

Training and evaluation. Every arm uses the same downstream encoder size, 300 pretraining epochs, batch size 32, and seeds $\{0, 1, 2\}$. The pretraining predictors, masks, and losses follow each

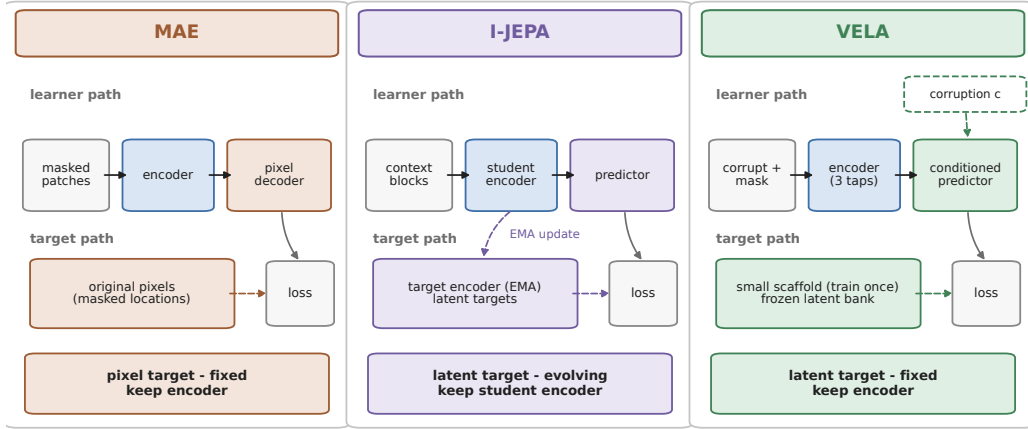


Figure 2: Target pathways. MAE reconstructs fixed pixels; I-JEPA predicts evolving EMA-teacher latents; VELA predicts fixed scaffold latents while conditioning its predictor on the applied corruptions. Dashed arrows denote target-side or conditioning paths.

Table 1: DeepLenseSim Model I $\rightarrow$ Model II. Values are mean $\pm$ sample standard deviation over three seeds. Best and second-best values are marked in bold and underline.

Method	Frozen 1%	Frozen 10%	Frozen 100%	Mass R^2
SIGReg-JEPA	.367 $\pm$.013	.458 $\pm$.023	.609 $\pm$.017	-.093 $\pm$.096
I-JEPA	.367 $\pm$.009	.505 $\pm$.023	.686 $\pm$.008	.164 $\pm$.065
I-JEPA (context-matched)	.343 $\pm$.003	.511 $\pm$.059	.697 $\pm$.050	.127 $\pm$.287
MAE	.412 $\pm$.003	.699 $\pm$.005	.832 $\pm$.015	.696 $\pm$.014
VELA (low-pass)	<u>.514$\pm$.041</u>	<u>.811$\pm$.009</u>	.903$\pm$.007	.817 $\pm$.008
VELA (bands)	.531$\pm$.036	.815$\pm$.011	<u>.900$\pm$.003</u>	.821$\pm$.012

method. Frozen features concatenate mean and maximum token pooling. We fit logistic regression using 100%, 10%, or 1% of the training labels, corresponding to 5,760, 576, and 57 examples, and evaluate on the fixed test split. Axion-mass results use ridge regression on an 80/20 split of the axion subset. End-to-end fine-tuning selects a checkpoint on validation accuracy and evaluates the test set once.

Baselines and masking audit. MAE reconstructs pixels with a 75% mask. I-JEPA uses the released-style four target blocks of 15–20% area and an EMA teacher. This leaves a median of 24 visible tokens, compared with VELA’s 50. Context-matched I-JEPA means that its masking configuration is changed so the encoder sees approximately the same number of visible tokens as VELA; using two target blocks leaves a median of 49. SIGReg-JEPA replaces the EMA teacher with the isotropic regularizer from LeJEPA [3]; it is an adaptation to this masked single-view harness, not a reproduction of the published multi-view recipe.

4 Results

4.1 Representation quality and label efficiency

Both full VELA variants produce the strongest frozen features in Table 1. Relative to MAE, VELA with band targets improves full-label accuracy by 6.87 percentage points, 10%-label accuracy by 11.69 points, 1%-label accuracy by 11.92 points, and mass regression by 0.125 in R^2 . Each difference has the same sign at seeds 0, 1, and 2. End-to-end accuracy follows the same, smaller ordering: 0.882 ± 0.002 for VELA-bands, 0.875 ± 0.009 for VELA-low-pass, and 0.861 ± 0.003 for MAE.

Matching I-JEPA’s context budget does not close the gap. VELA-bands exceeds context-matched I-JEPA by 18.81, 30.43, and 20.39 points at 1%, 10%, and 100% labels, respectively, and by 6.15

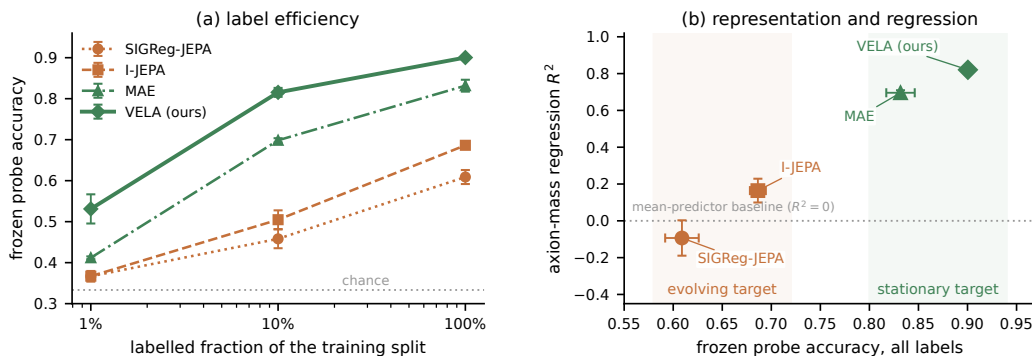


Figure 3: Frozen evaluation on DeepLenseSim. (a) Accuracy as the labelled fraction is reduced. (b) Full-label accuracy and axion-mass regression. Error bars show one sample standard deviation across three training seeds. The plot summarizes an empirical grouping in this benchmark; it does not by itself establish target stationarity as the causal mechanism.

points after fine-tuning. The matched arm is close to the released-style arm despite observing about twice as many tokens. Thus the original context imbalance was a real protocol defect, but it does not account for the repaired VELA result.

4.2 What the controls resolve

Low-pass and residual-band targets are effectively tied: their full-label accuracy differs by 0.30 points, their 10%-label accuracy by 0.41 points, and their R^2 by 0.004. The experiment therefore does not support the stronger hypothesis that information-disjoint residual bands are necessary for the downstream gains.

The scaffold and depth controls show a different pattern. With an untrained frozen scaffold, seeds 0 and 2 reach full-label accuracies of 0.893 and 0.898, but seed 1 falls to 0.608 and has chance-level end-to-end accuracy. Supervising only the deepest encoder block gives 0.892 and 0.891 in seeds 0 and 1, but 0.663 in seed 2 and negative mass R^2 . In contrast, all six runs of the two full VELA variants remain near 0.90. With only three seeds, these outcomes suggest—but do not establish—that scaffold training and supervision at several depths mainly improve reliability.

5 Discussion

The repaired experiment supports a practical conclusion: on this small scientific-imaging benchmark, VELA’s fixed latent targets yield more transferable frozen features than the evaluated JEPA and pixel-reconstruction baselines. The largest gains appear when downstream labels are scarce, which is the regime motivating self-supervision for strong lenses.

The mechanism is less settled than the headline numbers. Stationary targets are consistent with the observed ordering, but MAE and VELA differ in more than target motion, and VELA includes conditioning and auxiliary losses that have not been ablated here. The band decomposition is not load-bearing at this resolution, while the trained scaffold and depth ladder appear to reduce seed failures rather than raise the best-seed result.

The study is limited to one simulation suite, one split, three seeds, and a shallow encoder; its uncertainty estimates cover training variation, not new simulators or observing conditions. We also do not test composition or inversion of the corruption-conditioned operator. Larger encoders, additional instruments, and direct operator tests are needed before broader claims.

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